ACME Org — Employee Salary Management System

Software Requirements • Assessment Scope

1. Goal

Build a web application for a single HR Manager to manage employee records and annual gross base salaries for 10,000 employees across multiple countries. Replace spreadsheet-based workflows with searchable records, efficient pagination, and clear salary summaries.

2. Scope and Core Features

- **Employee management:** Create, view, update, and delete records containing employee ID, name, department, designation, country, employment status, annual gross base salary, and salary currency.
- **Salary dashboard:** Show headcount and the average, minimum, and maximum annual gross base salary per currency. Filter by department, country, and employment status, or break down by department or country. Never combine currencies.
- **Employee directory:** Search by name or employee ID; filter by country, department, employment status, and currency, with optional minimum and maximum salary (only together with a currency); use backend pagination of 20 records per page.
- **Data integrity:** Validate required fields, unique employee IDs, non-negative salary values, and supported currency codes on the backend.
- **Seed data and tests:** Provide a repeatable script for 10,000 fictional employees and automated tests for CRUD, validation, filtering, pagination, and dashboard calculations.
- **Technology:** FastAPI backend, React frontend using JavaScript/JSX, and a relational database (SQLite for local development, with PostgreSQL as the deployment target).

3. Deliberate Exclusions and Rationale

- **Authentication and advanced authorization:** One HR Manager persona is sufficient for the assessment. The public demo uses synthetic data only and is not intended for real employee information.
- **Payroll execution and payslips:** No net salary, taxes, deductions, allowances, bonuses, payment status, or bank transfers; the system manages salary records, not payroll execution.
- **Currency conversion:** Retain each salary's original currency. No cross-currency aggregation without a separately documented conversion method.
- **Salary history and audit logging:** Store current employee and salary information only to keep the initial implementation focused.
- **External integrations and statutory payroll compliance:** SSO, accounting integrations, approval workflows, and country-specific tax or employment rules are excluded because they require additional requirements and specialist validation.

4. Assumptions and Success Criteria

All seeded records are fictional. Salary is annual gross base salary in the recorded currency, with no additional compensation components. Success means the application supports 10,000 records, returns paginated results without loading the full dataset into the browser, calculates currency-specific summaries correctly, and passes automated tests. Deliver a public deployment, source repository with incremental Git commits, setup instructions, and a short demo video. This assessment demo is not a production payroll system.